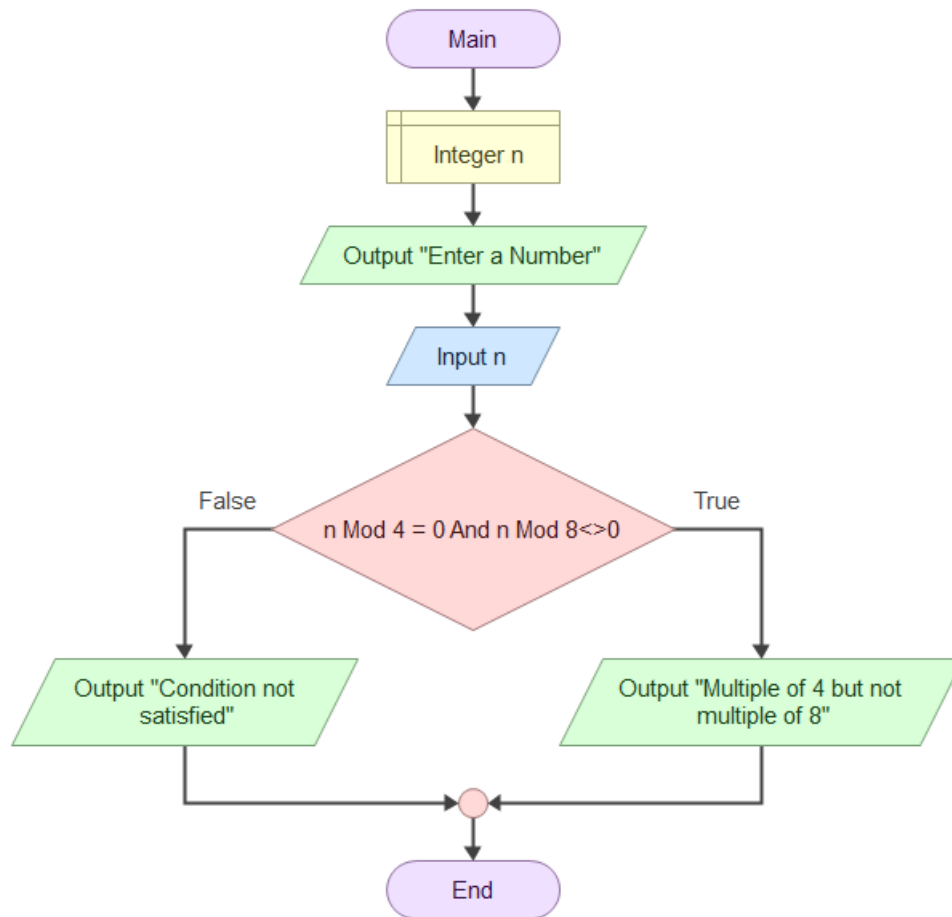


1. Read a number and determine whether it is a multiple of 4 but not a multiple of 8.

Flow Chart:



Output :

Enter a Number

Multiple of 4 but not multiple of 8

Psedocode :

Source Code Viewer

Auto Pseudocode

```
0  Function Main
1      Declare Integer n
2
3      Output "Enter a Number"
4      Input n
5      If n Mod 4 = 0 And n Mod 8 <> 0
6          Output "Multiple of 4 but not multiple of 8"
7      Else
8          Output "Condition not satisfied"
9      End
10 End
```

Java Code :

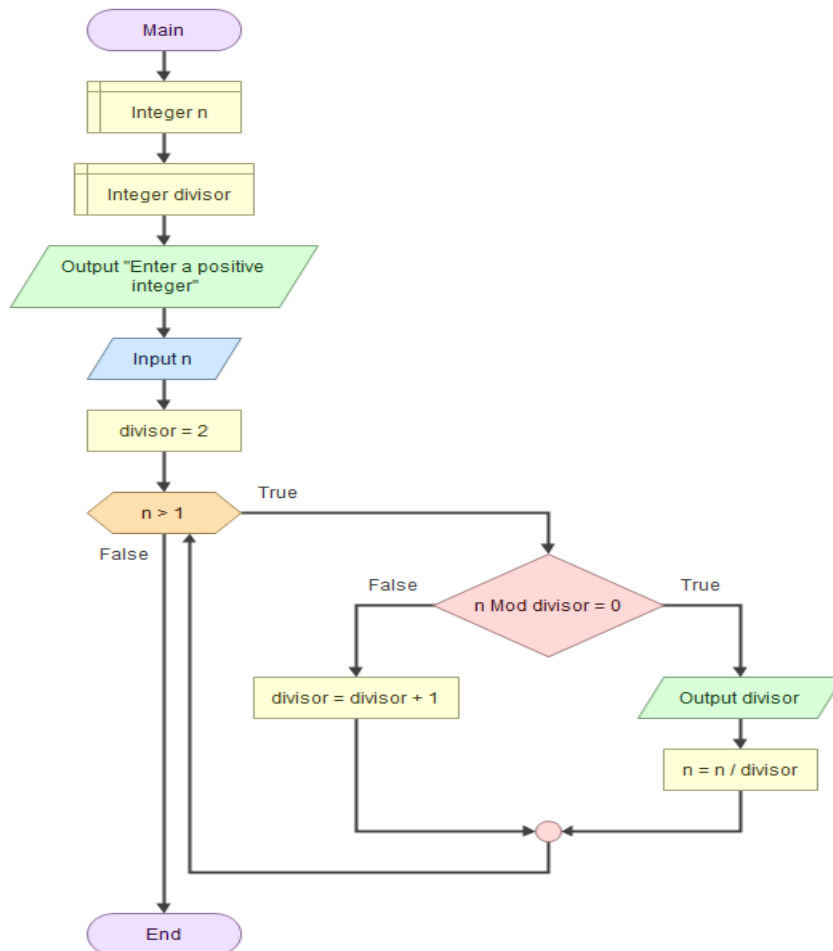
Source Code Viewer

Java

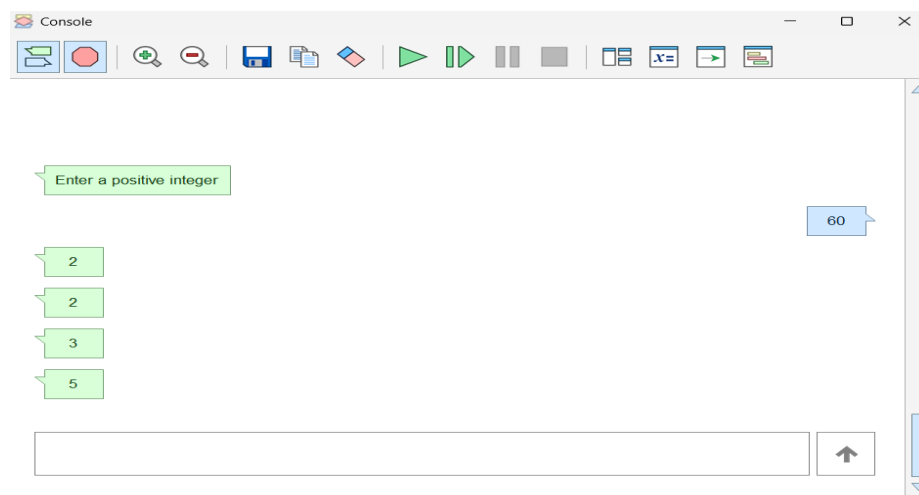
```
0  import java.util.*;
1  import java.lang.Math;
2
3  public class Program {
4      private static Scanner input = new Scanner(System.in);
5
6      public static void main(String[] args) {
7          int n;
8
9          System.out.println("Enter a Number");
10         n = input.nextInt();
11         if (n % 4 == 0 && n % 8 != 0) {
12             System.out.println("Multiple of 4 but not multiple of 8");
13         } else {
14             System.out.println("Condition not satisfied");
15         }
16     }
17 }
```

2. Read a positive integer and display its prime factorisation using repeated division.

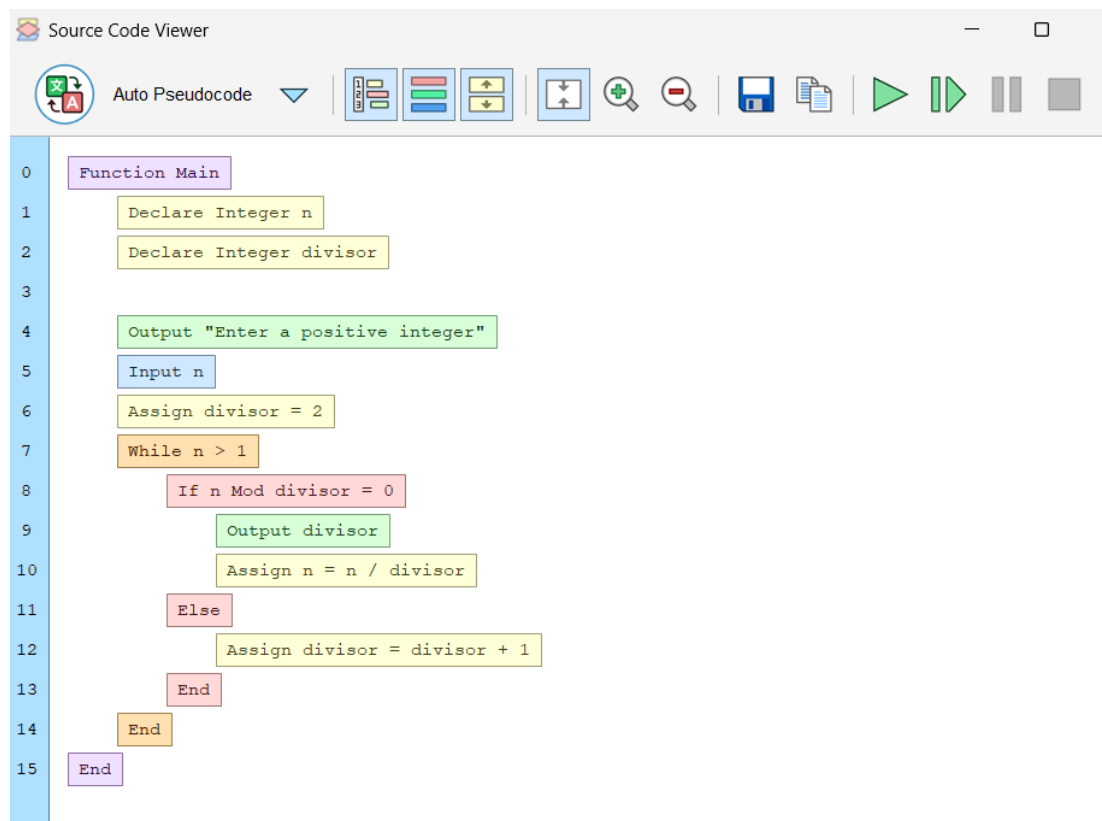
Flow Chart:



Output :



Psedocode :



Java Code :

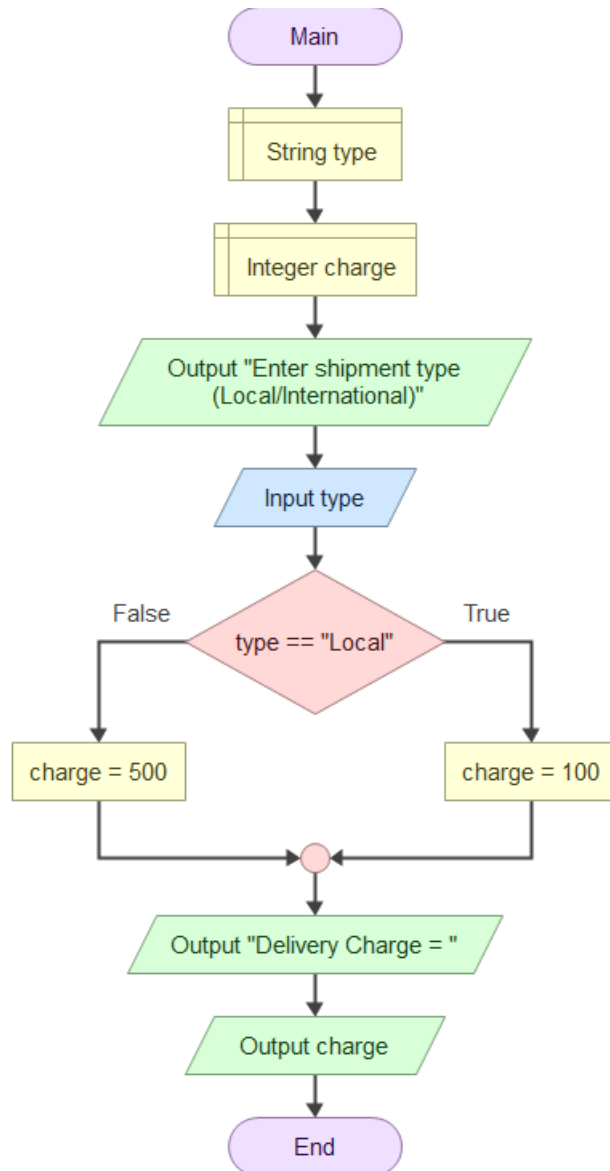
Source Code Viewer

Java

```
1  import java.lang.Math;
2
3  public class Program {
4      private static Scanner input = new Scanner(System.in);
5
6      public static void main(String[] args) {
7          int n;
8          int divisor;
9
10         System.out.println("Enter a positive integer");
11         n = input.nextInt();
12         divisor = 2;
13         while (n > 1) {
14             if (n % divisor == 0) {
15                 System.out.println(divisor);
16                 n = (double) n / divisor;
17             } else {
18                 divisor = divisor + 1;
19             }
20         }
21     }
22 }
```

3.Design a package-delivery flowchart with local and international charges. Explain how Java packages could organise classes such as customer, shipment, and billing.

Flow Chart :



Output:

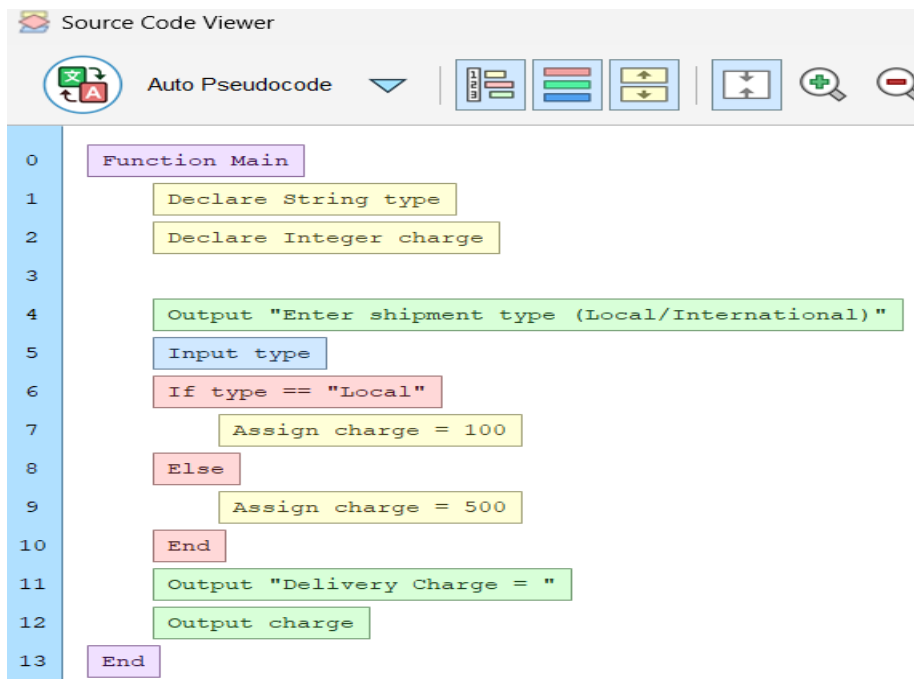
Enter shipment type (Local/International)

Local

Delivery Charge =

100

Pseudocode:



Java Code:

Source Code Viewer

```
4     private static Scanner input = new Scanner(System.in);
5
6     public static void main(String[] args) {
7         String type;
8         int charge;
9
10        System.out.println("Enter shipment type (Local/International)");
11        type = input.nextLine();
12        if (type.equals("Local")) {
13            charge = 100;
14        } else {
15            charge = 500;
16        }
17        System.out.println("Delivery Charge = ");
18        System.out.println(charge);
19    }
20 }
```